

Barney Jin

peiyuan3@illinois.edu | <https://realbj-dot.github.io/BarneyJin-Portfolio/> | +1 (317) 316-7876

EDUCATION

UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN

Jan 2024 — May 2025

Master of Computer Science (GPA: 3.8/4.0)

Champaign, IL

Core Coursework: Generative AI for Software Engineering, Operating System Design, Cloud Networking

UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN

Aug 2019 — May 2023

Bachelor of Science in Mathematics and Computer Science; Minor in Statistics (GPA: 3.6/4.0)

Champaign, IL

Core Coursework: Data Structures, Data Analytics, Algorithms, Linear Algebra, Database Design, Computer System Organization

SKILLS

Programming Languages: Python, Java, TypeScript, JavaScript, C++, C, SQL, HTML, R, Golang, Haskell, Dart, Swift

Tools & Frameworks: Flutter, SwiftUI, UIKit, React/React Native, Node, NextJS, Prisma, MongoDB, Firestore, Firebase, AWS,

Jenkins, Docker, Terraform, PyTorch, TensorFlow, Figma

PROFESSIONAL EXPERIENCES

Advanced Cad Cam Service (EngineeringPeople)

Sep 2024 — Present

Lead Software Engineer

- Architected NextFanUp (NFU), a live sports statistics and rewards platform, integrating **Rolling Insights** and **Game Sport Odds APIs** to ingest real-time scores, player stats, and betting odds data across professional leagues.
- Designed a dual-layer caching strategy locally and server-side with optimized **Time-To-Live (TTL)** structural configurations, significantly slashing external API cost metrics and minimizing redundant network requests.
- Engineered a transactional ecosystem leveraging **Cloud Firestore** and **Firebase Storage** to power peer-to-peer user betting lines, monetary reward mechanics, dynamic social feeds, and real-time friendship system graphs.
- Served as Lead Engineer for 'Go Birdie Go', delivering a cross-platform golfing analytics application using **Flutter (Dart)** and Material Design to ensure seamless adaptive rendering across iOS and Android platforms.
- Implemented cloud serverless architectures via **Python** and **Firebase Cloud Functions** to automate premium subscription lifecycle engines and secure transactional authentication workflows.

John Deere

July 2023 — Jan 2024

Software Engineer | Skills Tracking Application (Tracktor)

- Standardized the deployment of an internal skills-tracking application by implementing **Infrastructure as Code (Terraform)**, eliminating manual server configurations and ensuring strict cross-environment consistency.
- Engineered a fully automated **CI/CD pipeline** utilizing **Jenkins** and **AWS** cloud infrastructure, optimizing deployment turnaround times for core internal tools from hours down to minutes.

John Deere

Dec 2022 — July 2023

Full-stack Developer | Security Access Control (SAC)

- Developed a critical internal security portal built on **Next.js**, replacing legacy paper-based workflows to yield a cost reduction of over \$30k and recover 175 operational hours annually.
- Engineered high-performance state management systems utilizing **TanStack Query** and **Jotai**, verifying real-time concurrent data synchronization across comprehensive multi-step permission forms.
- Designed structured database schemas via **Prisma ORM**, implementing batch selection mechanisms that accelerated permission audit timelines for compliance administrators.

O2Micro International Limited

Wuhan, China | Jun 2021 — Aug 2021

Data Science Intern

- Developed automated data extraction scripts using **Python** and **Pandas**, replacing legacy workflows to accelerate physical battery performance lifecycle analysis arrays.
- Implemented **Principal Component Analysis (PCA) algorithms** to uncover primary variance dependencies in multi-dimensional experimental testing datasets, systematically isolating statistical background noise.
- Optimized experimental parameters based on analytical takeaways, refining internal execution workflows to drive a 25% efficiency increase across engineering team testing operations.

PROJECTS

SmartSync Reminder App (iOS) | Swift, SwiftUI, SwiftData, OpenAI API. Built a native iOS productivity application using **SwiftUI**, enforcing decoupled **MVVM software architecture** practices and integrating **EventKit API** for bi-directional **Apple Calendar** synchronization.

Real-time Web Content Filter | Python, JavaScript. Engineered a lightweight URL analysis engine and runtime browser extension using a custom **Python** classification algorithm to process and flag explicit content metadata patterns instantly.

Social Book Discovery Platform | Next.js, MongoDB, Python. Architected a full-stack reading management platform using **Next.js server-side rendering** to lower latency, backed by optimized **MongoDB** schemas to map complex multi-user relational states.